

NumPy Interview Questions (30)

1. What is NumPy?
2. Why is NumPy faster than Python lists?
3. What is an ndarray in NumPy?
4. What is the difference between a Python list and a NumPy array?
5. How do you create a NumPy array?
6. What is the difference between `np.arange()` and `np.linspace()`?
7. What are the different ways to initialize arrays in NumPy?
8. How do you check the shape, size, dimensions, and data type of an array?
9. What are NumPy data types (dtype)?
10. How do you change the data type of a NumPy array?
11. What is array indexing in NumPy?
12. What is slicing in NumPy?
13. What is Boolean indexing?
14. What is Fancy indexing?
15. What is broadcasting in NumPy?
16. What are the broadcasting rules in NumPy?
17. What is vectorization, and why is it important?
18. What is the difference between `reshape()` and `resize()`?
19. What is the difference between `flatten()` and `ravel()`?
20. What is the difference between a view and a copy in NumPy?
21. How do you concatenate multiple arrays in NumPy?
22. How do you split an array into multiple sub-arrays?
23. What are Universal Functions (ufuncs) in NumPy?
24. What is the purpose of the axis parameter in NumPy functions?
25. What is the difference between `axis=0` and `axis=1`?
26. What is the difference between element-wise multiplication and matrix multiplication?
27. What is the difference between `np.dot()`, `@`, and `np.matmul()`?
28. How do you generate random numbers in NumPy, and what is the purpose of `np.random.seed()`?
29. How do you find unique values, counts, and frequencies in a NumPy array?
30. What are the best practices for optimizing NumPy code performance?